

Cryptocurrency Liquidity Prediction - Final Report

1. Problem Statement

Predict the liquidity ratio of cryptocurrencies using market data to assess market stability.

2. Dataset

Source: CoinGecko

Files: coin_gecko_2022-03-16.csv and coin_gecko_2022-03-17.csv

3. Preprocessing

- Merged both days' data
- Handled nulls using backfill
- Feature engineered:
 - price_ma_5
 - price_volatility_5
 - vol_mcap_ratio
 - price_change_rate

4. Models Used

Model	RMSE	MAE	R ²
Linear Regression	0.329	0.112	-0.064
Random Forest	0.189	0.036	0.647
XGBoost	0.201	0.033	0.601

Selected Model: Random Forest 

5. Streamlit Deployment

- Input: 9 market features
- Output: Predicted liquidity ratio
- Live frontend for users via app.py

6. Screenshots



Cryptocurrency Liquidity Prediction

Fill in the market details below to predict the **Liquidity Ratio** of a cryptocurrency.

Current Price (USD)

150.3200

- +

Price Change 1h (%)

0.4500

- +

Price Change 24h (%)

2.1000

- +

Price Change 7d (%)

-5.7000

- +

24h Volume (USD)

200000000.0000

- +

Market Cap (USD)

10500000000.0000

- +

5-Day Moving Avg

148.7600

- +

5-Day Volatility

4.2000

- +

Abs 24h Price Change (%)

3.5000

- +

Predict Liquidity

 Predicted Liquidity Ratio: 0.481097

7. Conclusion

This end-to-end ML project includes preprocessing, modeling, and deployment — making it ideal for production or academic use.